

Technology Stack

Date	15 March 2026	Team ID	Not provided
Project Name	AI StudyBuddy	Maximum Marks	2 Marks

Define Your Technology Stack:

The following technology stack is derived from the supplied AI StudyBuddy backend code. Only technologies explicitly visible in the code are listed as selected technologies. Where the reference template asks for a technology that is not shown in the supplied code, it is marked "Not specified in supplied code" rather than being assumed.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	Not specified in supplied code	The provided code contains the backend/API implementation only. A frontend framework such as React, Angular, or plain HTML/CSS is not shown in the supplied code.
2	Backend / Server-Side	Node.js + Express.js	The server is created with Express and uses CommonJS modules. Express handles JSON middleware, CORS, API route mounting, the health-check endpoint, and global error handling. The default server port is 5000.
3	Database / Data Storage	MongoDB + Mongoose	Mongoose is used to connect to MongoDB through MONGO_URI and define User and Material schemas/models. MongoDB stores users and study materials including summaries, flashcards, quizzes, and study plans.
4	Cloud / Hosting / Deployment	Not specified in supplied code	The code shows a local/default Express configuration with PORT=5000 but does not identify AWS, Azure, Heroku, Docker, or another hosting/deployment platform.
5	Version Control & CI/CD	Not specified in supplied code	No GitHub, GitLab, Jenkins, GitHub Actions, or other version-control/CI/CD configuration is present in the supplied code.
6	Third-Party APIs / Other Tools	Google Gemini Generative AI Multer JWT bcryptjs CORS	Gemini is used for AI content generation through @google/generative-ai. Multer handles file uploads and restricts uploads to .txt, .md, and .pdf with a 5 MB limit. jsonwebtoken provides JWT verification, bcryptjs hashes and compares passwords, and cors controls cross-origin access.

Technology details visible in the supplied code

- Node.js dependencies shown: express, express-async-errors, cors, mongoose, dotenv, jsonwebtoken, bcryptjs, multer, and @google/generative-ai.
- Environment variables referenced: MONGO_URI, JWT_ACCESS_SECRET, GEMINI_API_KEY, CLIENT_URL, and PORT.
- AI model configured in the supplied Gemini utility: "gemini-3.7-flash".
- Data models shown: User and Material.
- Material data includes title, content, filename, summary, flashcards, quiz, and studyPlan.
- Authentication uses Bearer JWT tokens and an admin role check.